

Vision — Full System Architecture

How the platform is structured, how the backend and AI agent work, how research vs farmer data are used, and how grazing decisions (Good / Watch / High concern) are produced.

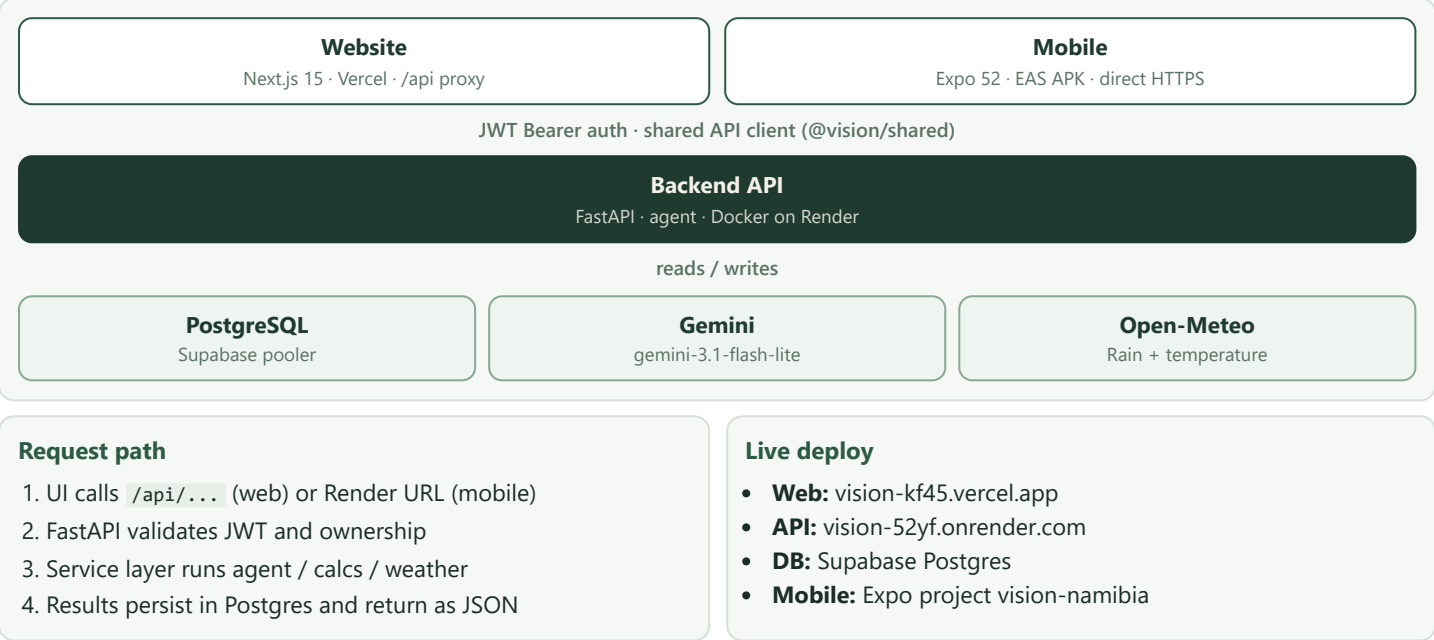
FastAPI + Gemini tool-calling

Rule-based fallback

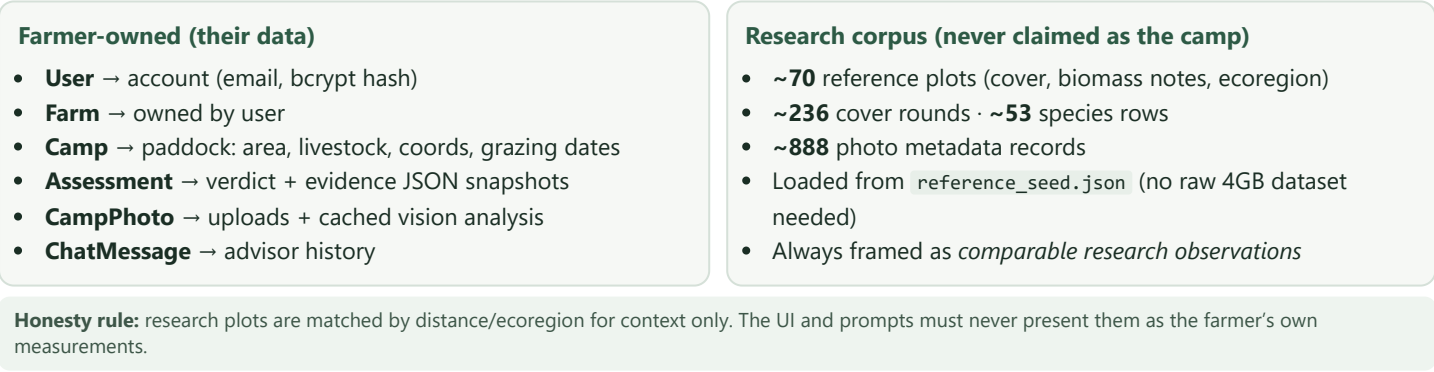
Supabase · Render · Vercel · EAS

Namibia rangeland research seed

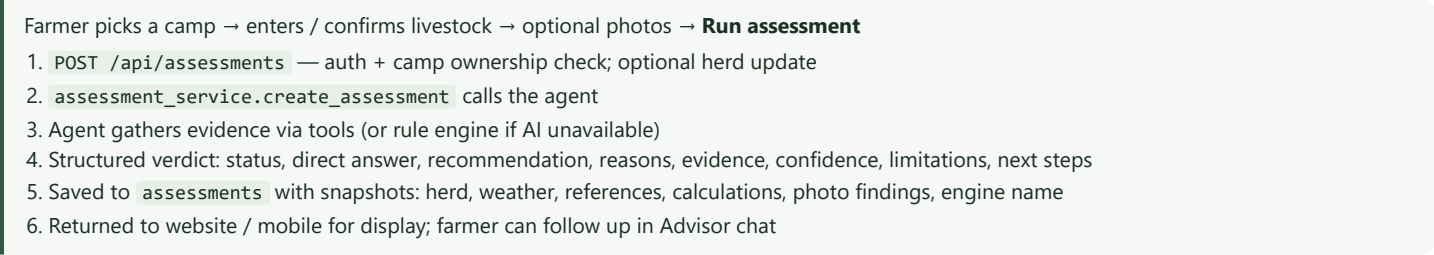
1. System topology



2. Data model — farmer vs research



3. End-to-end assessment flow



4. How the AI agent works

Tool-calling loop

- Model: Gemini via OpenAI-compatible SDK
- Up to **8** tool rounds (`MAX_TOOL_ROUNDS`)
- `tool_choice=auto` ; last round **forces** `submit_assessment`
- Final output is a function call — not free-form JSON
- Statuses: Good · Watch · High concern · Insufficient data
- Confidence: Low · Moderate · High

Prompt framing

- Separate farmer data, weather, research, photos, calcs, AI conclusions
- Cautious language (“may indicate”, “based on available evidence”)
- Photos never invent exact biomass / grass % / grazing days left
- Missing weather: say so and continue
- Low confidence → recommend field inspection / extension officer

5. Agent tools (evidence sources)

Tool	What it does	Source label
get_farm_and_camp	Farm + camp herd, area, coords, grazing	farmer-provided
list_camps	All camps on a farm	farmer-provided
search_dataset	Free-text / ecoregion research search	historical dataset (research plots)
find_comparable_plots	Top similar plots (distance + ecoregion)	comparable research plots
check_rainfall	7 / 14 / 30-day past + 7-day forecast rain	live weather (Open-Meteo)
check_weather_context	Temperature + drought hint from 30d rain	live weather (Open-Meteo)
calculate_livestock_per_hectare	LSU and stocking density	deterministic calculation
calculate_grazing_duration	Days since grazing start	deterministic calculation
get_previous_assessments	Last 5 assessments for the camp	previous assessments
compare_camps	Side-by-side camps on a farm	mixed sources
analyze_photos	Vision buckets (density, soil, dryness...)	visual photo observations
submit_assessment	Final structured verdict (assessment only)	—

Matching: haversine distance; -400 score bonus for ecoregion match; labels like “geographically nearby” (≤60 km). LSU: 1 cattle = 1.0 LSU; 6 goats/sheep = 1.0 LSU (Namibian extension convention).

6. How decisions are made

Good

Camp can currently sustain the herd under available evidence.

High concern

Strong overgrazing / drought stress signals — act soon.

Watch

Early pressure or decline — monitor and plan changes.

Insufficient data

Not enough evidence to judge responsibly.

A) Gemini path (preferred)

1. Model chooses which tools to call
2. Builds evidence items labelled by source
3. Calls submit_assessment with status + confidence
4. Backend overlays tool snapshots onto the saved row
If the loop ends without submit, or the LLM errors → fallback.

B) Rule-based fallback (always available)

Used when no API key, LLM fails, or tools never submit.

- Always computes LSU/ha, grazing days, weather, comparable plots (+ photos if any)
- Adds a **concern** score from thresholds below
- Maps score → status with template recommendations

7. Fallback scoring rules (transparent thresholds)

Livestock pressure

- LSU/ha ≥ 0.25 → concern +2
- LSU/ha ≥ 0.12 → concern +1
- Below that → low pressure
- Missing area → limitation (no LSU/ha)

Rainfall (30-day)

- < 10 mm → +2 (or +1 if no herd)
- < 40 mm → +1 (if herd)
- Else → reasonable rainfall reason
- Weather unavailable → continue, note it

Plots & photos

- Avg grass cover of refs < 25% → +1
- Photo grass = sparse → +1
- Photo exposed soil = high → +1

Status mapping from concern score

- No known signals → **Insufficient data** (Low confidence)
- No herd (rested): concern ≥ 2 → Watch, else Good
- With herd: concern ≥ 4 → **High concern**; ≥ 2 → **Watch**; else **Good**
- Next steps always include walk-the-camp; extension officer if Watch / High concern with a herd

8. Photos, chat, and offline clients

Photos (vision)

- Optional; strengthen, never replace other evidence
- Buckets only: grass density, dryness, exposed soil, woody density, erosion signs
- Analysis cached on `CampPhoto`
- No LLM → all “uncertain”

Advisor chat

- `POST /api/chat` — same tools, no `submit_assessment`
- Free-text reply; last 8 history turns
- Needs Gemini (no rule-engine verdict)
- Persists user + assistant messages

Offline (shared client)

- Cache farms, camps, past assessments
- Queue camp create/update offline
- Online-only: login, assess, chat, photos, weather
- Read timeout ~18s · write ~90s

9. Tech stack & tools summary

Layer	Stack / tools
API	Python 3.12, FastAPI, Uvicorn, SQLAlchemy 2, Alembic, Pydantic, JWT + bcrypt, pandas, Pillow, httpx, OpenAI SDK
AI	Google Gemini <code>gemini-3.1-flash-lite</code> (tool calling + vision); deterministic fallback engine
Weather	Open-Meteo (cached ~1h, never blocks assessment on failure)
Database	PostgreSQL via Supabase session pooler
Website	Next.js 15, React 19, TypeScript, Tailwind, Leaflet, Recharts
Mobile	Expo Router, Secure Store, AsyncStorage, NetInfo, Image Picker, Location, Speech
Shared	<code>@vision/shared</code> — API client, offline cache/queue, design tokens
Deploy	GitHub · Supabase · Render (Docker) · Vercel · Expo EAS
Auth	HS256 JWT (~14 days); demo <code>demo@vision.na</code> / <code>vision123</code>

10. Backend modules (where the logic lives)

API

- `api/auth.py` · `farms.py` · `camps.py`
- `api/assessments.py` · `chat.py` · `compare.py`
- `api/photos.py` · `dataset.py`

Agent & services

- `agent/agent.py` · `tools.py` · `prompts.py` · `fallback.py`
- `services/assessment_service.py` · `matching_service.py`
- `calc_service.py` · `weather_service.py` · `llm.py` · `photo_service.py`